

Cleansing Oil^{*}

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Abstract

I study the dynamics of greenhouse gas emissions following disruptions in crude oil markets. Following a 10% unexpected increase in crude oil prices, CO₂-equivalent atmospheric concentration declines by 3.24% relative to trend after two years. The decline is heterogeneous across greenhouse gases and reflects a reshuffling of fossil fuel use rather than a transition toward cleaner energy. Natural gas and petroleum products decline substantially, while lignite and bioenergy generation increase. Consistent with tighter financial conditions and elevated uncertainty, solar capacity also contracts following the shock. I find no evidence of accelerated adoption of alternative-fuel vehicles. Overall, the results show that an oil shock provides some of the incentives for decarbonization associated with a carbon tax, but also generates distortions that a carbon tax does not.

1 Introduction

Oil supply disruptions are widely believed to accelerate the green transition and slow down global warming. Whenever geopolitical tensions threaten global crude oil supplies, it is routinely argued that higher fossil fuel prices will encourage the adoption of cleaner technologies and reduce greenhouse gas emissions. This paper shows that this view is only partially true.

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I find that a 10% increase in oil prices caused by an exogenous oil supply shock reduces global greenhouse-gas concentrations and increases emissions efficiency. However, the adjustment occurs mostly because of a decrease in energy consumption, and less from the substitution away from dirty and into green energy. Oil supply disruptions reduce emissions, but they do not accelerate the clean-energy transition, neither in the private vehicle market, nor in the power sector, at least in the short to medium run.

Oil supply disruptions trigger a broad reallocation across fuels rather than a simple substitution away from fossil energy. Consumption of petroleum products falls sharply, especially jet fuel, gasoline, and diesel. Natural gas consumption declines as well. At the same time, lignite production and bioenergy generation increase. These adjustments generate heterogeneous greenhouse-gas dynamics. Atmospheric concentrations of carbon dioxide and methane decline, whereas nitrous oxide concentrations rise persistently. The reduction in overall warming potential therefore masks important changes in the composition of emissions and in the underlying energy mix.

The contraction in renewable deployment is particularly surprising. Conventional reasoning suggests that higher fossil-fuel prices should improve the competitiveness of clean alternatives and stimulate their adoption. Indeed, much of the discussion surrounding oil disruptions implicitly assumes that expensive oil acts as a catalyst for the green transition. Yet oil supply shocks are not carbon taxes. The same disturbance that raises the price of fossil fuels also depresses economic activity, increases uncertainty, tightens financial conditions, and disrupts complementary energy markets. These forces work against renewable investment and can dominate the standard relative-price channel, which is what I find.

The distinction matters for how oil disruptions should be viewed through a climate lens. An oil shock delivers a partial carbon-price signal, but it also generates distortions that a carbon tax does not. The resulting environmental consequences are therefore more nuanced than commonly assumed. Oil-intensive activities contract, yet renewable deployment weakens and electricity generation shifts toward lower-quality combustion alternatives such as lignite and bioenergy. Moreover, electricity reliance — as a secondary energy source — does not increase.

To study these effects, I bring the empirical tools of modern macroeconomics to the study of climate change and energy transitions. Combining identified oil supply shocks with global monthly data on greenhouse-gas concentrations, fossil-fuel consumption, electricity generation, renewable capacity, and vehicle registrations, I estimate the dynamic responses of emissions and energy use to exogenous disruptions in oil markets. The analysis spans the period from 2001 to 2025 and traces how oil supply shocks propagate through the energy system — from fuel consumption and electricity generation to technology adoption, and ultimately global greenhouse-gas concentrations.

1.1 Related Literature

This paper contributes to five strands of literature. First, it relates to the large literature studying the macroeconomic consequences and transmission channels of oil-market shocks. Beginning with Hamilton (1983), a vast literature has documented that oil-price increases are associated with declines in economic activity and changes in inflation, investment, and financial conditions. More recent contributions have emphasized the importance of distinguishing between different sources of oil-price fluctuations. For example, Kilian (2009) shows that supply, aggregate-demand, and precautionary-demand shocks have markedly different macroeconomic consequences, while Baumeister and Hamilton (2019) develop a Bayesian framework to incorporate prior knowledge about the different disturbances. Building on techniques typically used in the analysis of monetary policy, Känzig (2021) uses high-frequency information around OPEC announcements to identify exogenous oil-supply news shocks and study their macroeconomic effects. I leverage this literature for identification, but shift the focus towards the environmental and energy consequences of oil shocks.

Second, the paper contributes to the literature studying the response of fossil-fuel demand to energy price changes. Much of the attention in this literature has been devoted to gasoline markets — in particular, to estimating the price elasticity of gasoline demand. A growing body of work documents how this elasticity has evolved over time and varies across households and regions, with Kilian and Zhou (2024) providing a recent account using cross-state variation in oil-to-gasoline pass-through as a new instrument. Far less attention has been devoted to other petroleum derivatives and primary fuels: diesel, jet fuel, liquefied petroleum gas (LPG), natural gas, and coal each face distinct demand structures and substitution margins that a gasoline-centric lens cannot capture. The importance of this broader coverage is underscored by recent events: the threatened closure of the Strait of Hormuz in 2025 raised acute concerns about the supply of jet fuel for global aviation and bunker oil for international maritime trade. I contribute by documenting the response of the full fossil-fuel spectrum — gasoline, diesel, jet fuel, liquefied petroleum gas, natural gas, and coal — to exogenous oil-supply shocks.

Third, this paper relates to the literature studying the relationship between greenhouse-gas emissions and macroeconomic fluctuations. Heutel (2012) establishes that carbon emissions are deeply procyclical in the United States. Using a cross-section of countries and annual data, Doda (2014) confirms this procyclicality internationally and further documents that emissions are more volatile than output. Closer to my work, Khan et al. (2019) provide a first attempt at identifying the causal effects of different macroeconomic shocks — including oil-price and technology shocks — on US emissions, but do not find statistically significant effects for any of them. I document significant effects of oil-supply shocks on emissions, with the added value of conducting a world-wide analysis — important for a global problem like climate change.

Fourth, the paper contributes to the macroeconomic literature on energy efficiency, directed technical change, and capital structures. In their foundational work, Hassler, Krusell, and Olovsson (2021) show that the short-run substitution elasticity between energy and capital/labor is virtually zero (near-Leontief), but that sustained energy price increases direct innovation toward energy-saving technologies, rendering long-run substitutability significantly higher. Using time-series methods, Känzig and Williamson (2024) isolate these technological drivers, finding that while energy price shocks successfully stimulate energy-saving innovation, they are highly recessionary and crowd out other productive innovations. Conversely, pure energy-saving technology shocks drive a non-recessionary decoupling of output and emissions. Furthermore, Casey (2024) studies optimal taxation in a model with energy use and directed technical change. He argues that a Cobb-Douglas production function drastically overstates the long-run emissions reductions of simple energy taxes due to rebound effects that discourage subsequent innovation on energy-use, emphasizing the need for targeted R&D policy. I complement this literature by studying whether oil-price shocks improve emissions efficiency at the global level and by tracing the underlying mechanisms through detailed adjustments in fossil-fuel use, transportation demand, and electricity generation. I show that while an oil-price shock improves emissions efficiency per unit of industrial production, this is a short-lived effect, driven mostly by adjustments in the intensive margin. It does not translate into sustained emissions efficiency gains, nor does it spur an energy transition towards renewable energy sources.

Fifth, this paper contributes to the literature studying the adoption of clean-energy technologies and the structural and financial frictions that govern the green transition. While conventional theory suggests that fossil-fuel price spikes accelerate renewable adoption, macro-financial and technological frictions can disrupt this channel. First, renewable projects are highly capital-intensive and face upfront expenditures, rendering them hypersensitive to the cost of capital (Bistline, Mehrotra, and Wolfram 2023). Inflationary energy price shocks trigger monetary policy tightening; because renewables are far more capital-intensive than their fossil-fuel substitutes such as natural gas, the resulting rise in the cost of capital creates a differential financial barrier to clean energy adoption. The financial headwinds extend to the innovation frontier: Aghion et al. (2024) document that contractionary monetary policy reduces green patenting significantly more than non-green patenting, as tight credit disproportionately constrains the young firms at the forefront of clean innovation. This contrasts with the finding of Känzig et al. (2025) that green innovation is broadly countercyclical — driven by a “green is in the future” channel that insulates long-term R&D from short-term recessions. Second, grid-level technological constraints create interdependencies across fuels. Verdolini, Vona, and Popp (2018) document a strong complementarity between intermittent wind and solar and fast-reacting fossil-based backup capacity (primarily natural gas). Consequently, when oil shocks simultaneously disrupt natural gas supply

and spike gas prices, the adoption of solar is severely hindered. Indeed, the relationship between natural gas and renewables is complex; while the shale gas boom reduced emissions by displacing coal, Acemoglu et al. (2023) and Harstad and Holtsmark (2026) argue that cheap natural gas can create a "fossil-fuel trap" that delays clean innovation, whereas Lindequist and Selent (2025) find evidence of gas acting as a green "bridge fuel". I contribute to this literature by documenting the negative effect of oil price shocks on clean energy adoption: both solar-based electricity and electric vehicle adoption decline following such events.

1.2 Organization

The remainder of this paper is organized as follows. First, section 2 studies the implications of oil price hikes on worldwide greenhouse gas concentrations. Section 3 documents the dynamics of a broad spectrum of fossil fuels. Section 4, in turn, details the mechanisms that underlie changes in emissions efficiency — particularly focusing on the extensive margin — the adoption of clean energy alternatives. Section 5 concludes.

2 Global Warming Potential of Oil Price Shocks

Oil price shocks affect greenhouse-gas emissions through various channels with potentially opposing forces. On the one hand, an increase in oil prices directly decreases demand, thereby lowering GHG emissions derived from petroleum consumption. On the other, it induces substitution dynamics, increasing the use of alternative sources, more and less polluting, such as coal and solar, respectively. The first counteracts the emissions decline from a fall in crude oil demand, the latter exacerbates it.

In this section I quantitatively evaluate the overall emissions effect of oil price shocks using standard macroeconomic time-series techniques, and thus the ensuing global warming potential changes they generate.

2.1 Data

Emissions I use monthly average global atmospheric concentration data from the National Oceanic and Atmospheric Administration (NOAA) to study the global warming effects of oil price shocks. NOAA provides measurements for 4 of the main greenhouse gases (GHG): carbon dioxide (CO_2), nitrous oxide (N_2O), methane (CH_4) and sulfur tetrafluoride (SF_6). The data start at different dates, so I restrict my sample to the intersection of their spans. As a result, all data start in January 2001 and end in December 2025. Starting the sample in the early 21st century has the advantage of coinciding with the rise of renewables as viable energy sources, a significant development for the world economy's energy-source choice set.

In order to measure the global warming potential of the combined gas concentrations added, I compute the carbon dioxide equivalent (CO_2e) for the 4 GHGs for which I have data. This individual index measures how much CO_2 would be necessary to increase the globe’s temperature by the same amount as that resulting from the combined GHG emissions since pre-industrial times, all else equal, i.e., keeping the other GHG gases’ concentrations constant to pre-industrial levels. I follow the formulas and parameters provided by Etminan et al. (2016) to first measure the Stratospheric-temperature-adjusted radiative forcing (SARF). Then, I fix all gases to their pre-industrial levels according to Etminan et al. (2016), and solve for the necessary change in CO_2 to attain the same SARF. I provide more details on the parameters and algorithm used in the supplemental appendix (section B).

Oil Price Shocks To capture exogenous variations in the price of crude oil, I use the oil supply news shocks series provided by Känzig (2021). To generate this series they first exploit the institutional features of the Organization of Petroleum Exporting Countries (OPEC): at pre-determined dates OPEC systematically provides an outlook on their own crude oil production. Using high-frequency oil futures data, they can then measure the changes in the price of crude oil futures contracts in a narrow window of time around these scheduled events. This high-frequency approach thus avoids reverse causality problems and isolates supply from demand-driven fluctuations in oil price futures. In a second step, using a standard proxy vector autoregression, they are able to identify exogenous shifts in the price of crude oil due to changes in the expected supply of global crude oil by applying instrumental variables to the residuals of the oil supply equation and using the news surprise series previously generated. The resulting estimates are the exogenous shock series, used widely in the macroeconomics literature. The series is normalized so that a 1-unit shock corresponds to a 10% increase in the price of crude oil on impact.

Mori and Peersman (2024) document that this identification approach is subject to informational deficiencies: the identified structural innovations are found to Granger-caused by financial variables, indicating that markets can predict OPEC decisions through information not captured by the vector auto-regressive model, thereby contaminating the instrument. They show that augmenting the econometrician’s information set with financial indicators resolves these issues and yields sharper, more stable estimates. I address this concern directly by including financial controls in my local projections, as described in subsection 2.2.

In principle, the structural supply shocks of Baumeister and Hamilton (2019) would be better suited for my purposes: unlike news shocks, they are identified to produce an immediate fall in oil production, representing a cleaner physical supply disruption rather than a shift in expectations over future production, matching my motivation. However, over the sample, from 2001 to 2024, these shocks yield a positive response on industrial production regardless of the controls

included, contradicting both theory and the empirical literature, and indicating that they may also be capturing disruptions in the demand for oil. Thus, this renders them unsuitable for this analysis¹. The oil supply news shocks of Känzig (2021) do not suffer from this problem: they deliver the expected contractionary response, confirming that the identification remains valid over the sample period. Notwithstanding, they generate identical dynamics in the price of oil and most other macroeconomic indicators — the signals that agents respond to.

2.2 Methodology

To assess the effect of oil price shocks, I resort to the local projections methodology popularized by Jordà (2005). For $h = 0, \dots, H$, I estimate the long-differences model:

$$\Delta^{h+1}y_{t+h} = \alpha_h + \alpha_{m(t),h} + \beta_h \cdot s_t + \sum_{k=1}^{12} (\rho_h \Delta y_{t-k} + \Gamma'_h X_{t-k} + \tau_h s_{t-k}) \quad (1)$$

where α_h is an intercept, to account for the average growth at horizon h . $\alpha_{m(t),h}$ are month fixed-effects aimed at capturing the seasonality, patterns present in both greenhouse gas emissions and oil prices. $m(t)$ is the calendar month at date t . s_t is the exogenous oil price shock and X_{t-k} are control variables. $\Delta^{h+1}y_{t+h} = y_{t+h} - y_{t-1}$ is the difference between the dependent variable y at horizon h relative to the month before the shock $t - 1$, and is either the concentration of individual GHG in the atmosphere, or of my measure of equivalent CO_2 . Following Montiel Olea and Plagborg-Møller (2021), I include 12 lags of the dependent variable and compute confidence bands using a wild bootstrap based on Liu (1988). The former show that combining lags of the dependent variable with heteroskedasticity-robust standard errors is a sufficient approach for local projection inference. I rely on Mammen’s (Mammen 1993) distribution to sample the bootstrap multipliers, using $B = 999$ repetitions. Details on the algorithm and the multiplier distribution are provided in the supplemental appendix (section C).

As discussed in the previous subsection, Mori and Peersman (2024) show that financial variables Granger-cause the oil supply news shocks, contaminating the instrument through anticipation effects. I address this by including 12 lags of the shock itself and of 2 financial controls: the global financial cycle of Miranda-Agrippino and Rey (2020), which captures broad movements in global risk appetite and cross-border financial conditions, and the financial uncertainty index of Jurado, Ludvigson, and Ng (2015), which measures the unpredictability of financial variables, reflecting forward-looking information that market participants embed in asset prices ahead of OPEC decisions. I keep this parsimonious specification uniform across all empirical sections of the paper, including the panel analyses where the available time span is considerably shorter, so that results remain comparable throughout.

1. See the supplemental appendix (section A) for a full account of this phenomenon.

The sample includes the COVID pandemic period. To handle this, I follow Baumeister and Hamilton (2025) and drop the year of 2020 completely from the data. Because of the inclusion of lags, 2021 is also effectively dropped.

2.3 Results

2.3.1 The Macroeconomic Backdrop

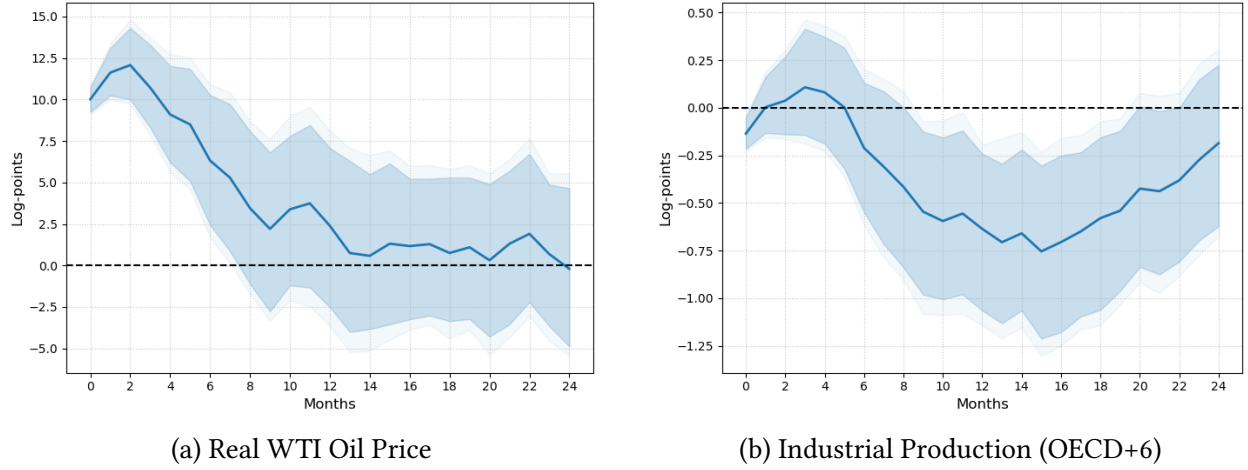
Before turning to the emissions results, figure 1 documents the response of key macroeconomic aggregates to the shock. This serves two purposes: it validates the shock identification for the restricted sample, 2001 to 2024 — excluding 2020, and it establishes the macroeconomic context within which all results should be interpreted. This exercise also helps to clarify our goal: we are not analyzing a pure decline in oil prices while everything else is kept constant — a useful theoretical question but one with less practical importance when the events that generate such oil price fluctuations also affect macroeconomic dynamics — even though they originate from the crude oil market alone. We are precisely interested in the joint dynamics induced by these events, coined "oil shocks".

Panel *a*) of Figure 1 shows the real oil price, computed by deflating the West Texas Intermediate price by the US consumer price index, both obtained from the Federal Reserve Economic Data (FRED) database. It rises by 10% on impact — by construction — and gradually decays, becoming statistically indistinguishable from zero at around 8 months, with point estimates centering at zero 1 year after — much less persistent than what Känzig (2021) finds. These price dynamics are very similar to those induced by the supply shocks of Baumeister and Hamilton (2019) over the same sample, so the implied price signals are closely comparable across the two series. Absent non-linearities introduced by supply disruptions, the two thus imply equivalent oil price signals to which economic agents react.

Panel *b*) of Figure 1 confirms that the shock is contractionary: industrial production for OECD and 6 selected countries², obtained from Baumeister and Hamilton (2019), falls on impact and remains depressed over the 2-year horizon, consistent with a negative aggregate supply shock. The drop peaks after 14 months, implying an almost 0.75% drop in global economic activity. The maximum effect is almost the same as in Känzig (2021). Distinctly, it is much less persistent, in line with the dynamics of the WTI.

2. Industrial production is used as a proxy for economic activity because it is available at the monthly frequency, contrarily to gross domestic product.

Figure 1: Macroeconomic Responses to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable in each panel is the $h + 1$ long-difference in the respective series. Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

2.3.2 Global Warming Potential

The backdrop surrounding oil news shocks has two effects on energy consumption: first, the slowdown in economic activity decreases overall demand for energy. Second, the increase in crude oil prices incentivizes substitution towards alternative energy sources. On the one hand, this effect can be negative, inducing more coal use — the most polluting fossil energy source. On the other, it can accelerate the uptake of renewable energy sources, such as solar. Naturally the horizons of these two substitution effects differ. The first could play out from the onset of the shock as coal plants with available capacity increase their output. The second can potentially take longer as it requires investments in new infrastructure since renewables tend to work at the limit of their useful capacity.

As a result, the net quantitative effect of oil price shocks on overall GHG emissions is not ex-ante determined. To assess it, panel *a*) of figure 2 presents the impulse response functions for the aggregate CO_2e index, estimated using equation (1). Following an oil supply news shock, atmospheric CO_2e declines by 0.11 parts per million (ppm) relative to trend after 4 months. Following a short rebound, the concentration decline persists and stabilizes at 0.10 ppm. This implies a short-term unitary semi-elasticity, and is statistical proof that increases in oil prices contribute positively to global cooling and lead to immediate decreases in global warming potential relative to trend. This decline is only marginal when compared to long-run trends, representing a 3.24 percentage point decline in atmospheric CO_2e concentration relative to its yearly average

increase, when measured over our sample, from 2001 to 2025.

While there is a decline in emissions, it may be a consequence of a decline in economic activity and not of increased dirty energy efficiency, i.e., in the amount of dirty energy necessary to support a certain amount of economic activity. To evaluate this, I compute the impulse response function for the logarithm of the ratio of the year-on-year difference in atmospheric CO_2e concentration over the industrial production index for OECD+ countries: $\log \left(\Delta^{12} CO_2e_t / IP_t \right)$. I use year-on-year differences instead of the month-on-month flow to deal with the seasonality in atmospheric GHG concentration.

The results are plotted in panel *b*) of figure 2 and show a fast but shortly-lived decline in emissions per unit of economic activity, mostly focused on months 3 to 6. Emissions efficiency fluctuates thereafter and drops again on the 14th month. The short-run dynamics in particular are similar if we instead use the yearly cumulative summation of industrial production as our denominator, as shown in the supplemental appendix (section D). Overall, these results indicate that the observed decline in emissions is not just a result of the ensuing economic slowdown but of increased emissions efficiency, in line with what Känzig and Williamson (2024) find for the United States. In contrast, I cannot statistically ascertain that the efficiency effect is permanent or even long-lived beyond the time oil prices mean-revert. This discrepancy may reflect sample selection. The shocks from Känzig 2021, when used in a longer sample used like in Känzig and Williamson (2024), are known to produce a more persistent decline in crude oil prices, which mechanically should sustain the efficiency incentive over a longer horizon.

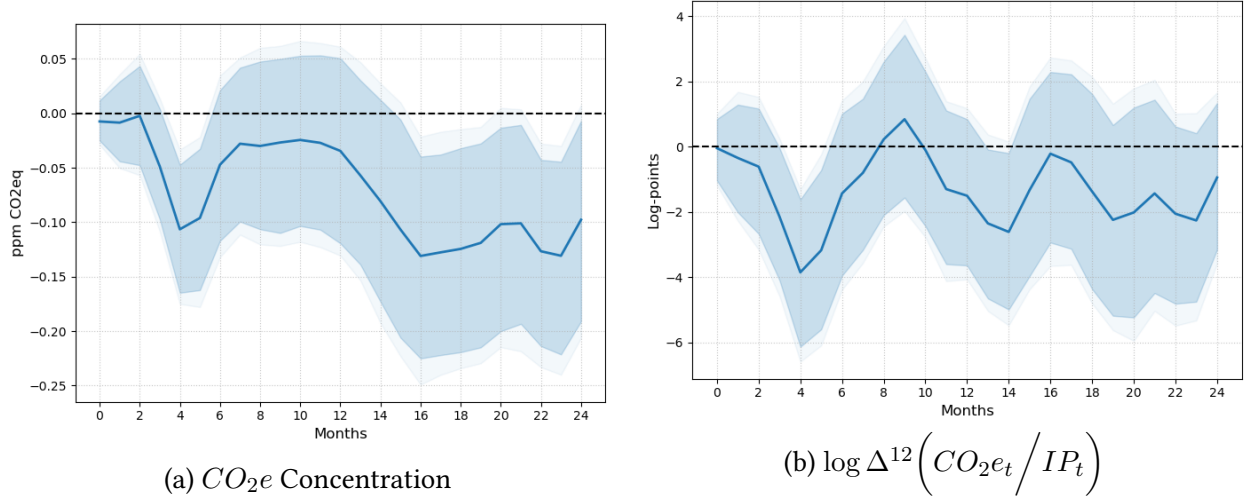
2.3.3 Individual Emissions

While overall global warming potential decreases as a result of an oil price shock, this behavior may not be homogeneous across different gases because of the ensuing energy mix reshuffling. Moreover, understanding the individual GHG dynamics is important because they have different atmospheric lifetimes, and so different global warming potentials depending on the horizon of analysis (Pachauri, Mayer, and Intergovernmental Panel on Climate Change 2015).

To verify this, I also study the dynamics of each individual GHG, plotted in figure 3. The results confirm the heterogeneous dynamics. The most abundant GHG in the atmosphere, CO_2 , follows a similar path to CO_2e : first it declines to 0.08 ppm and then plateaus at 0.10 ppm. Similarly, CH_4 decreases by 0.62 parts per billion (ppb) after 4 months but rebounds and its concentration mean-reverts, becoming statistically insignificant after 8 months.

Working in the opposite direction, the most powerful GHGs rise: SF_6 increases 4 months after the shock, peaking at 0.004 parts per trillion (ppt), but then reverting to the mean after 1 year. In the same tone, N_2O increases after 4 months, peaking at 0.04 ppb before fluctuating

Figure 2: Response of CO_2 -Equivalent Concentration to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. Panel (a): $h + 1$ long-difference in atmospheric CO_2e concentration (ppm). Panel (b): $h + 1$ long-difference in the log ratio of the 12-month difference in CO_2e concentration to the 12-month cumulative growth of industrial production. Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

down. In contrast, it remains elevated at around 0.05 ppb after 2 years. The persistent elevation of N_2O is consistent with the concurrent increases in lignite production and bioenergy generation documented in section 3 and subsection 4.2: both lignite combustion and biomass burning are potential sources of N_2O through incomplete high-temperature combustion.

The documented heterogeneous dynamics across GHGs motivate our study of hydrocarbons in section 3.

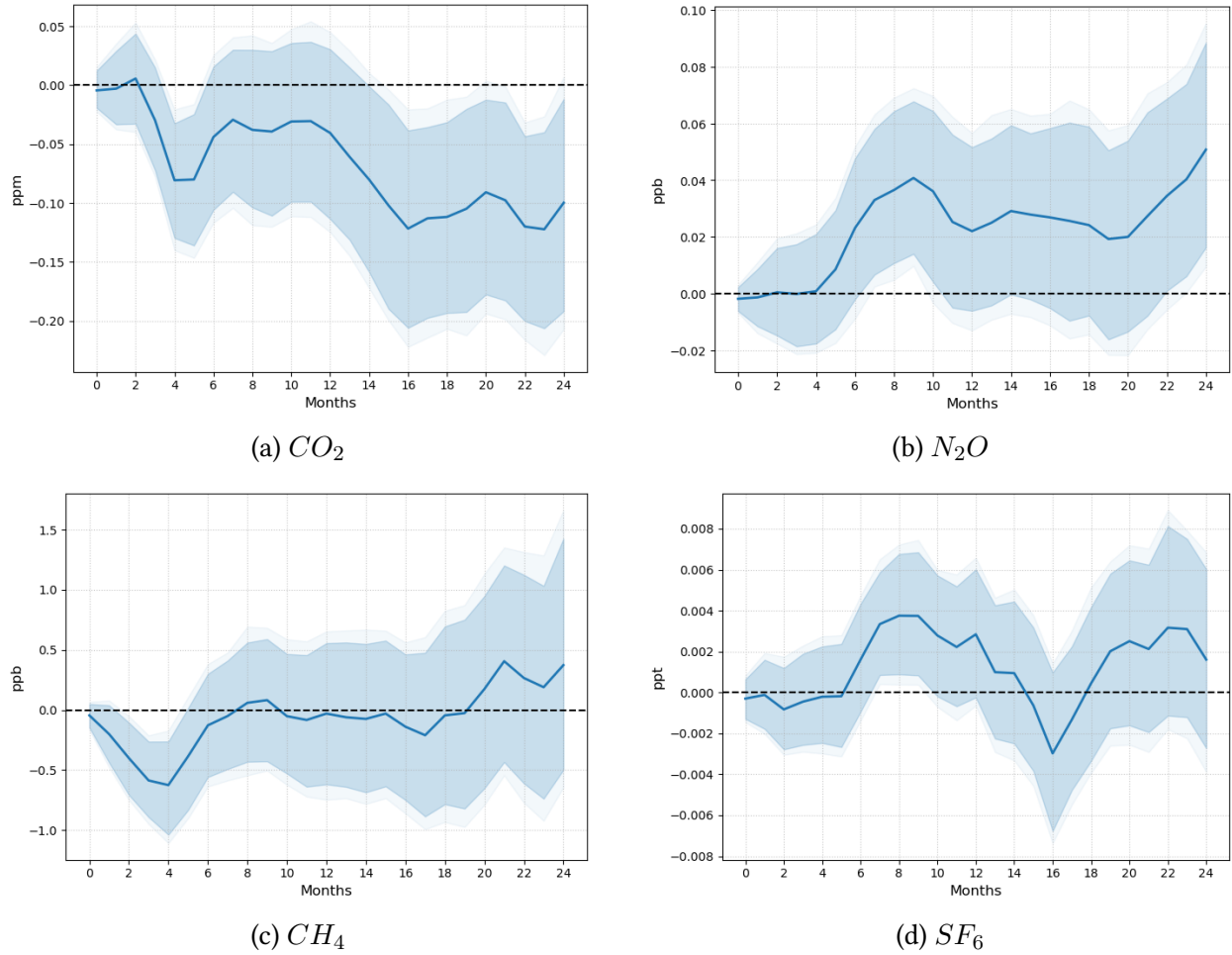
2.4 Robustness

I estimate several alternative specifications to ensure that our main results are robust. In particular, I deviate from the standard lag-length and consider including fewer (more) lags — for the lagged variables included in the right-hand side of equation (1) — to account for potentially shorter (longer) auto-correlation structures. I also exclude the oil news shock lags to show that the results are not purely driven by noise. Finally, to ensure that no outlier shocks drive the dynamics, I trim the shock to the 5th and 95th percentiles of the sample distribution, from 2001m1 to 2025m12, excluding any shocks outside this range from the estimation³.

I compare the alternative specifications' point estimates with our baseline irfs for CO_e and for the emissions-efficiency measure of figure 2 in figure 4 and confirm that the point estimates are

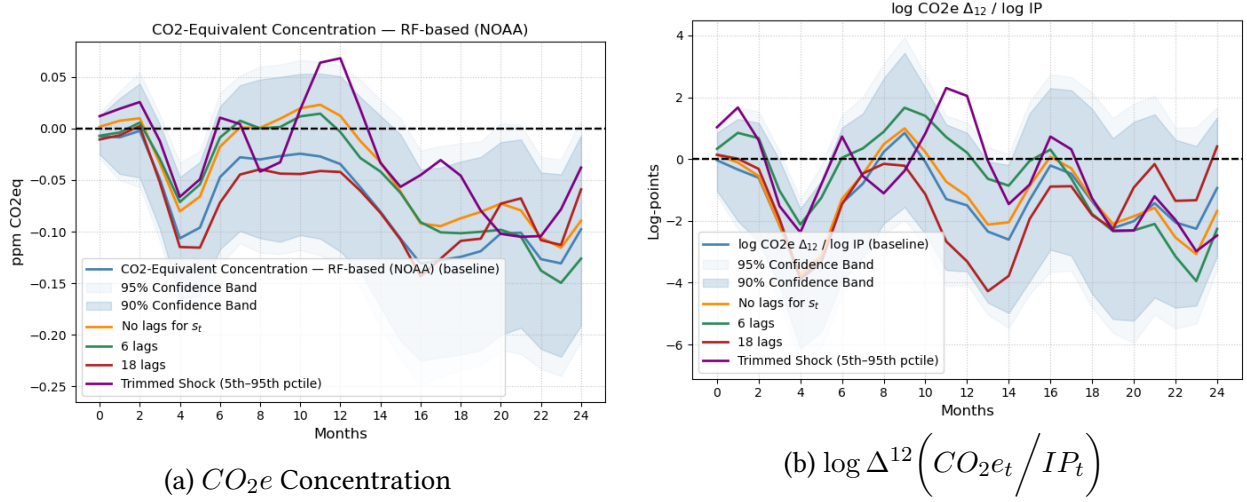
3. I keep the shock's lags unrestricted.

Figure 3: Response of Individual Greenhouse Gas Concentrations to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in atmospheric concentration. Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

Figure 4: Robustness of the CO_2e and Emissions-Efficiency Responses to Alternative Lag Specifications



Notes: Baseline impulse response (blue line, shaded 90%/95% confidence bands) to the oil supply news shock of Känzig (2021), overlaid with point estimates from three alternative lag specifications: no shock lags, 6- and 18 lags for all the controls. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

very similar across different specifications. The same comparisons are done for the other series with similar conclusions. Their plots are provided in figure D.1.

I also test for asymmetric dynamics, i.e., whether the impulse responses differ between expansionary and contractionary oil price shocks, by adding a new term $\gamma s_t \times 1_{\{s_t \geq 0\}}$. Figure D.2 in the supplemental appendix plots the resulting irfs for γ . The dynamics of CO_2e do not statistically differ from zero. In contrast, emissions efficiency, after 1.5 years, decreases more slowly following an expansionary oil shock, than it rebounds following a contractionary impulse.

3 Hydrocarbon Dynamics

In this section I study the sources of GHG that explain the concentration dynamics presented in section 2. I focus on primary dirty energy consumption. This largely includes industrial processes and energy production, which together represented about 84% of all GHG emissions worldwide (Climate Watch 2021). I thus exclude other sources of emissions like land-use, and agriculture. Specifically, I inspect the consumption and/or production dynamics of all types of fossil fuels following oil price hikes — less documented in the literature studying oil shocks. Overall, petroleum and natural gas consumption, and mainstream coal production decline while, dirtier less carbon-intense coal consumption and production temporarily rise.

3.1 Data

I build an unbalanced monthly panel dataset of countries from a variety of sources. All data are restricted to January 2001 onwards or later, depending on its availability⁴.

First, I obtain fuel consumption data from the Joint Organisations Data Initiative (JODI), a consortium that monitors and aggregates oil and gas data from national and international energy organizations. I obtain information on national refined oil fuel consumption for gasoline, diesel, jet fuel, residual oil — including bunker oil, mainly used for maritime shipping, and liquefied petroleum gas (LPG). These series thus cover primary energy derived from oil and gas (excluding electricity) used in the residential sector — specifically for heating and cooking — as well as the transportation sector, including automotive, aviation, and maritime transport. The data runs from January 2002 to December 2025 and covers more than 80 countries.

Secondly, I obtain natural gas consumption data, from the same data consortium, both for the whole economy and the electricity and heat generation sectors in particular. This dataset starts later, in January 2009, and includes more than 70 countries. JODI demand is a derived quantity: it is not directly measured but computed as a residual from the supply-side balance⁵. Errors in any balance component can therefore occasionally produce negative demand values, which I remove from the dataset. These observations would be excluded regardless, since all dependent variables enter in log-points.

Thirdly, in the absence of high-frequency consumption data, I obtain quarterly coal production data from the International Energy Agency. It has a much shorter time-span, starting in 2016, but covers a great share of the world’s production including China, Russia, and OECD countries. Although consumption and production of coal do not exactly match, like crude oil, their dynamics are similar — motivated by a desire for precautionary stockpiles.

3.2 Methodology

Ideally, I would build aggregate measures for each of the variables from the data collected. Nevertheless, the staggered inclusion of countries prevents such an approach. A uniform subsample would either not be entirely representative of global activity or would have a much shorter time-span. Additionally, a panel set-up can increase the statistical power of my estimates in place of shorter series as is the case for coal, natural gas or for the power sector data. As a result, I use the individual country observations in the estimation. Following Chodorow-Reich, Nenov, and Simsek (2021) and Ottonello and Winberry (2020), I adapt the methodology from subsection 2.2

4. Additional details about each dataset, its coverage and the raw time-series are provided in the supplemental appendix (section E).

5. Specifically: $\text{Demand} = \text{Refinery Output} + \text{Receipts} + \text{Imports} - \text{Exports} - \text{Products Transferred} + \text{Interproduct Transfers} - \text{Stock Change} - \text{Statistical Difference}$.

to the panel set-up by running a fixed-effect version of local projections:

$$\Delta^{h+1}y_{i,t+h} = \alpha_{i,h} + \alpha_{i,m(t),h} + \beta_h s_t + \sum_{k=1}^{12} (\rho_h \Delta y_{i,t-k} + \Gamma'_h X_{t-k} + \kappa_h \Delta C_{i,t-k} + \tau_h s_{t-k}) \quad (2)$$

where i indexes the country. $\alpha_{i,h}$ are country-specific fixed-effects to account for differential growth trends. $\alpha_{i,m(t),h}$ are country-specific month fixed-effects to capture the inherent seasonality of the dependent variable which may vary by country, especially because of different climates and geographies. X_{t-k} are the same global controls included in equation (1), and $\Delta C_{i,t-k}$ are lags of the logarithmic growth rate of country i 's nominal exchange rate in USD from the International Monetary Fund. This country-specific control partially accounts for time variation in countries' economic and financial conditions and has a very broad coverage — ensuring that no observations are lost. All the dependent variables are defined in log-points.

All regressions are estimated via weighted least squares, so that β_h can be interpreted as a representative global estimate Ramey 2019. We use the lagged raw dependent variable as weight, unless otherwise specified. Compared to static weights, using time-varying weights accounts for differential trends across countries' energy use patterns. Using a yearly moving average instead, to decrease the influence of seasonality, does not materially change the results — as shown in the supplemental appendix (subsection F.5). We always present the pre-trends with our panel impulse responses to ensure that the results are not arising from pre-existing dynamics that we fail to capture.

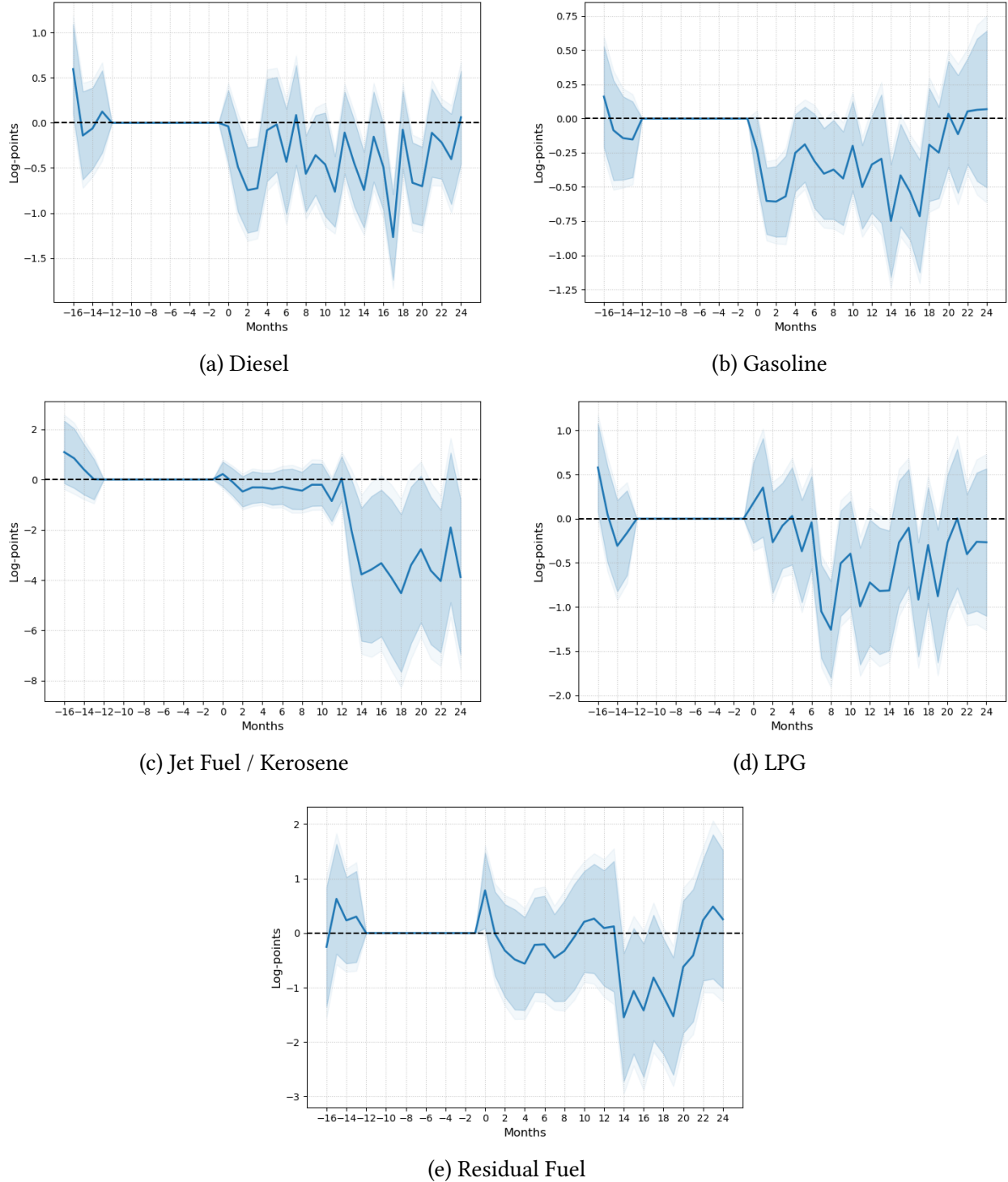
Following Almuzara and Sancibrian (2026), I cluster standard errors by time period: every country is exposed to the same aggregate oil shock at each date, which creates cross-sectional dependence across countries within a period. As in the time-series regressions, the lags of the dependent variable are included in equation (2) to capture the potential autocorrelation.

3.3 Refined Oil

I first plot the impulse response functions for refined oil products consumption in figure 5. All regressions are weighted by the lag of the respective raw petroleum variable. Consumption drops overall, albeit at different moments in time. Well documented in the literature, retail fuels, diesel and gasoline, drop almost instantaneously. In contrast, and less studied, residual fuel, liquefied petroleum gas, and jet kerosene decline only later. The first declines after 6 months, and the latter two types 1 year after the shock hits.

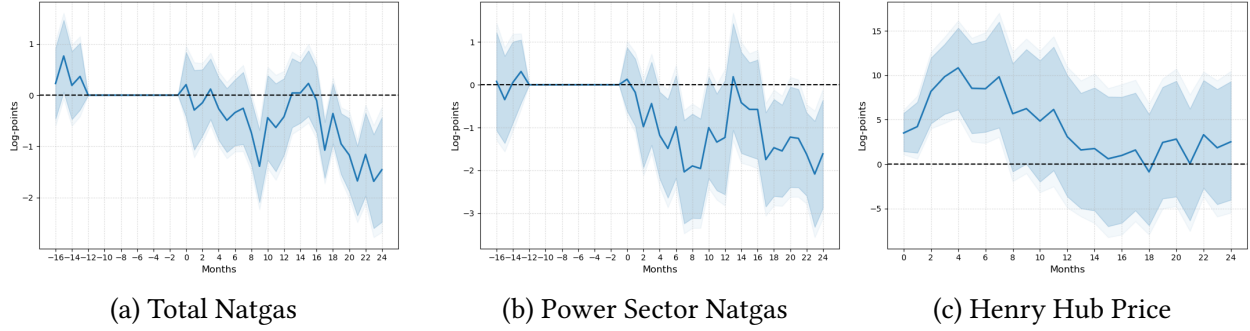
Their semi-elasticities to crude oil prices differ too. Diesel and gasoline decline and stay persistently around 0.61 percentage points below trend following an oil shock that raises crude oil prices by 10% on impact. LPG declines by 1.05 percentage points by month 7, while residual fuel decreases by 1.55 percentage points by month 14. All revert to the trend after 2 years. Differ-

Figure 5: Response of Refined Oil Consumption to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in the log of monthly national consumption (thousand barrels). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective consumption series (thousand barrels). COVID-19 (2020) is excluded from the sample. Shocks: January 2003 – December 2024; outcome data extends to December 2025.

Figure 6: Response of Natural Gas Consumption to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. Panels (a)–(b): the dependent variable is the $h + 1$ long-difference in log monthly national natural gas consumption (m^3). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective natural gas consumption series (m^3). COVID-19 (2020) is excluded from the sample. Shocks: January 2010 – December 2024; outcome data extends to December 2025.

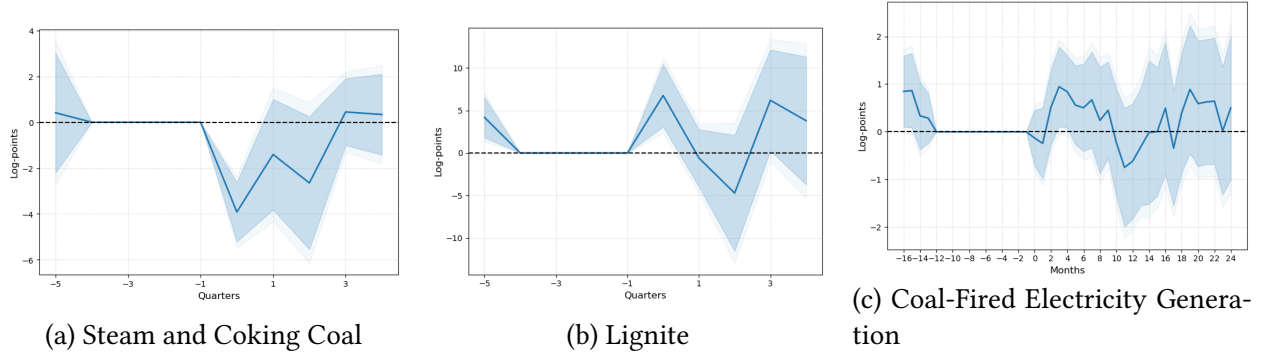
ently, jet fuel sees the biggest drop of 3.77 percentage points, only after 14 months, and does not recover within the horizon considered.

3.4 Natural Gas

Natural gas is a partial substitute for oil in power generation and industrial heating, so an oil price increase may divert demand toward gas (Baumeister et al. 2024). At the same time, oil and gas are typically extracted from the same wells (U.S. Energy Information Administration 2013). As a result, disruptions in the supply of the former are known to have consequences on the supply of the latter. Moreover, decreased economic activity decreases the demand for natural gas.

To determine which channel dominates, we can compare the dynamics of consumption and prices. Figure 6 presents the impulse response functions for natural gas consumption following an oil supply news shock. Natural gas declines in general, and in the power sector specifically. Total consumption continuously decays without recovery within the 2 years, dropping 1.46 percentage points by the 2-year horizon. The drop is more severe in the power sector, which declines by 1.61 percentage points relative to trend — also without rebound. This drop in natural gas consumption is followed by an increase in gas prices. The Henry Hub reference gas price, obtained from FRED, plotted in panel c) of figure 6 rises by 3.51 percentage points on impact, reaching a peak of 11 percentage points after 4 months before slowly declining. This suggests that adverse supply effects dominate both the demand-side substitution and income effects, explaining the dynamics we observe in consumption.

Figure 7: Response of Coal Production and Coal-Fired Electricity Generation to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. Panels (a)–(b): the dependent variable is the $h + 1$ long-difference in log quarterly coal production; the horizon is measured in quarters. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-quarter. Regressions are weighted by the lag of the raw production level. COVID-19 (2020) is excluded from the sample. Shocks: January 2017 – December 2024; outcome data extends to July 2025. Panel (c): the dependent variable is the $h + 1$ long-difference in log monthly coal-fired electricity generation (TWh). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of coal-fired electricity generation (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

3.5 Coal

Coal is the biggest source of pollution in primary energy production. It is a critical input in industrial production and a strong substitute for natural gas in power generation. To understand its dynamics following an oil price shock, I adapt the methodology presented in subsection 3.2 to the data’s quarterly frequency. Because of the different uses, I separate the main results by type of coal: steam and coking, and lignite, and leave the results for total coal production in the supplemental appendix (subsection F.1). Although steam coal is used primarily for electricity and heating, and coking coal is primarily used in an industrial setting — mainly metallurgy, the data reports their production jointly, so I cannot assess their dynamics separately.

Panels *a)* and *b)* of subsection 3.5 present the impulse response functions for each type of coal. I consider only a 1-year horizon since the sample is much shorter, starting mostly in 2016, and smaller, numbering only 30 countries. Steam and coking coal decrease by 3.92 percentage points on impact, but recover after 3 quarters. In contrast, lignite, the lowest ranked coal, increases by 6.76 percentage points on impact before reverting back to the mean. Overall, total coal production follows the dynamics of steam and coking coal, and decreases by -3.18 on impact, staying depressed for 6 months before recovering.

Because I am unable to separate coking from steam coal, I cannot draw conclusions regarding coal-based electricity production from these figures. To ascertain its net response, panel *c)* of

subsection 3.5 plots instead the monthly impulse response functions for coal-based electricity production⁶. It shows evidence of a temporary rise in the first 3 months, before mean-reverting, in line with the lignite dynamics. This indicates that the latter’s increase compensates for the potential fall in steam coal, and helps explain why N_2O emissions rise, as shown in subsection 2.3. Lignite can produce 17% to 80% more nitrous oxide than bituminous coal⁷ (Government of Canada 2024). As such, while we do not observe a sustained increase in coal-based electricity production, the change in the underlying coal mix contributes to the rise in N_2O -emissions intensity.

3.5.1 Bioenergy

Bioenergy — electricity generated from burning organic material such as wood pellets, agricultural residues, and dedicated energy crops — occupies an intermediate position between fossil fuels and clean renewables. Unlike solar and wind, it is dispatchable and emits GHGs during combustion, including CO_2 and N_2O .

Figure 8 presents the impulse response of bioenergy generation. Following the shock, bioenergy rises persistently, reaching 0.83 percentage points at the 2-year horizon. Though the bands are wider, certain horizons are statistically significant at conventional levels, confirming the elevated trend in biomass usage. This is consistent with the substitution dynamics in the dispatchable segment of power generation, as the prices of coal and natural gas rise. This increase in biomass combustion further contributes to the persistent elevation of atmospheric N_2O concentration documented in section 2.

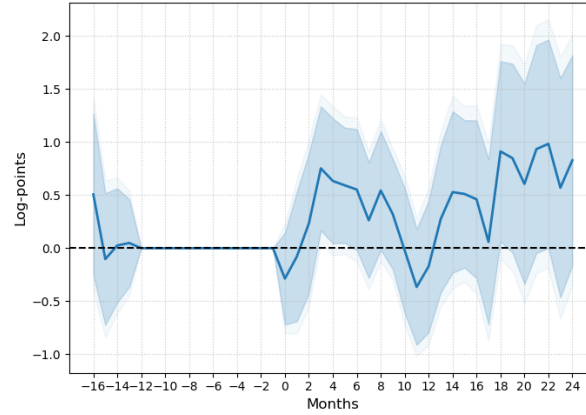
4 Jump-Starting The Energy Transition

The increase in oil and gas prices is typically viewed as an opportunity for electrification and for the adoption of alternative energy sources, in particular, renewable energy sources. In this section, I study those dynamics. Specifically, I focus on the behavior of (private) road transportation, the biggest petroleum consuming sector (International Energy Agency 2023), and electricity generation. Currently, these are the two major energy-consumption activities with the widest and most effective set of alternative technologies — to fossil fuels — available. Notwithstanding, I find no evidence of an acceleration towards cleaner energy alternatives as a result of oil shocks — in line with the purely temporary increase in emissions-efficiency documented in subsubsection 2.3.2.

6. Details about the data are presented in subsection 4.2.

7. Government of Canada 2024 estimate that lignite produces 2 g of N_2O per GJ of energy generated when used for utility level electricity whereas sub-bituminous and bituminous produce 1.1 to 1.7 g/GJ.

Figure 8: Response of Bioenergy Generation to an Oil Supply News Shock (TWh)



(a) Bioenergy Generation (TWh)

Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly bioenergy generation (TWh). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of bioenergy generation (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

4.1 Road Transportation

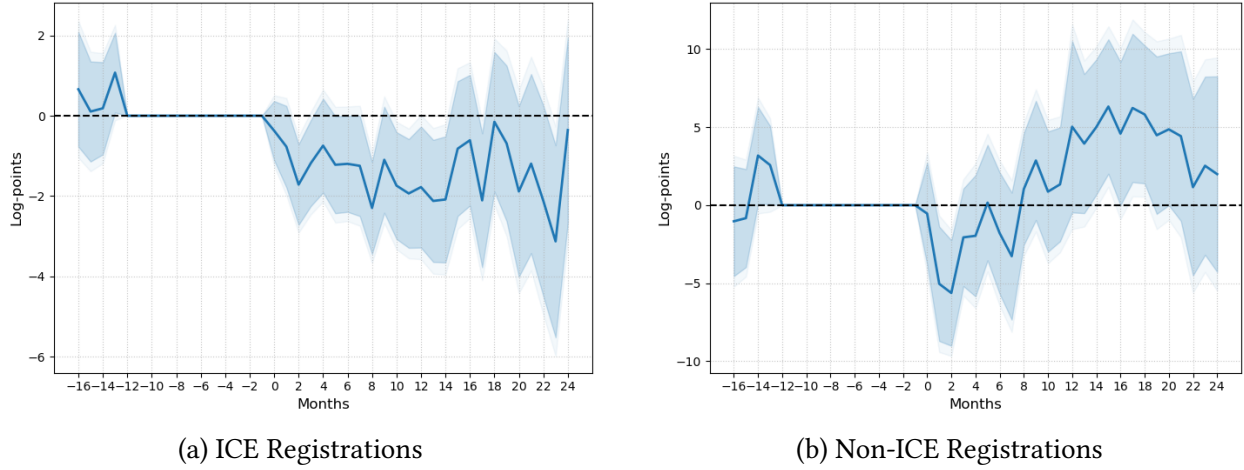
I focus on the private vehicle market. On top of data availability, personal lower-emission vehicles are much more developed than alternative transportation sectors, such as aviation or shipping, and so can be considered viable alternatives to internal combustion engine (ICE) cars.

To study how non-ICE vehicles' attractiveness changes following increases in oil prices, I use vehicle registration data from Andrew (2026), who collects monthly new registrations from national statistical registries. This dataset starts in January 2001 and covers more than 80 countries. I repeat the monthly panel methodology described in subsection 3.2 and start the sample in 2001 to be in line with the remaining exercises.

Figure 9 plots the impulse responses of standard ICE registrations alongside all non-ICE registrations — a composite category including zero-emission vehicles (ZEV) — battery electric and hydrogen — plug-in and non-plug-in hybrids, and alternative-fuel ICE vehicles such as LPG and flex-fuel cars. ICE registrations fall clearly and persistently, reaching a maximum decline of 2.30 percentage points 8 months after the shock and remaining depressed over the 2-year horizon, consistent with consumers delaying or cancelling purchases of high-operating-cost vehicles.

Non-ICE registrations display similar dynamics at first, with registrations falling by -5.63 after 2 months. As we show in the supplemental appendix (subsection F.4) this decline happens across zero-emission and hybrid vehicles, but does not affect alternative-fuel ICE vehicles, such as LPG and flex-fuel cars. As a result, the percentage of new non-ICE registrations actually falls at first.

Figure 9: Response of Vehicle Registrations to an Oil Supply News Shock



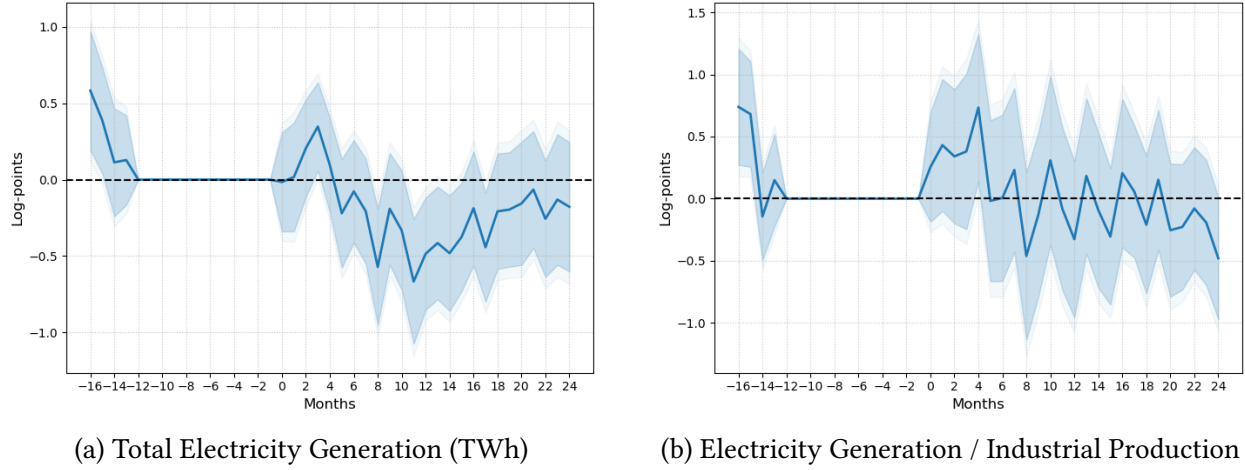
Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly vehicle registrations. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective vehicle registration count. Non-ICE includes zero-emission vehicles (battery electric, hydrogen), plug-in and non-plug-in hybrids, and alternative-fuel ICE vehicles (LPG, flex-fuel). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025. Total registrations (ICE and non-ICE combined) are reported separately in the supplemental appendix (figure F.5).

After an initial fall, non-ICE registrations recover and rise above trend, reaching a peak of 6.32 after 15 months. In contrast, this overshoot is driven mostly by ZEVs and hybrids, while alternative-fuel ICE cars increase faster, starting 6 months after the shock, but also mean-revert earlier — while the former two categories continue to increase. This rebound is actually not sufficiently large for us to observe a statistically significant increase in the percentage of non-ICE registrations — so this means that the rebound just follows the dynamics of overall vehicle purchases; it does not represent an accelerated transition towards alternative energy sources in private road vehicles.

4.2 The Power Sector

Two opposing forces shape the electricity sector’s response to an oil supply shock. First, the contraction in economic activity documented above suppresses aggregate energy demand, pulling total electricity generation down. Second, higher oil prices compress the competitive advantage of oil-intensive processes and raise the relative attractiveness of electrification and renewable generation in particular. These substitution incentives operate at two margins: the intensive margin, where existing power plants run closer to installed capacity, and the extensive margin, where higher expected returns accelerate investment in new infrastructure. Section 3 highlighted the former, whereby lignite coal production rises and natural gas consumption de-

Figure 10: Response of Total Electricity Generation to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. Panel (a): the dependent variable is the $h + 1$ long-difference in log monthly total electricity generation (TWh). Panel (b): the dependent variable is the $h + 1$ long-difference in the log ratio of monthly electricity generation to the industrial production index, normalised to a common 2019m12 base. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Panel (a) is weighted by the lag of total electricity generation (TWh); panel (b) is weighted by the previous year's real GDP. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

creases in the power sector. This section focuses on the latter.

I obtain data on electricity generation by source, and wind and solar installed capacity come from Ember, a think tank that compiles data from national and international energy organizations. I obtain monthly TWh of electricity generated from coal, gas and other fossil fuels, nuclear, wind, solar, hydro, and other renewable sources, covering more than 80 countries.⁸ The data start in January 2001, but the bulk of countries enter between 2010 and 2015. Capacity data record GW of nameplate capacity for wind and solar installations – including privately owned rooftop solar – starting only in 2016 and covering about 25 countries. Because this data does not account for depreciation and has such short and narrow coverage, we only use it as a robustness check, and instead focus on electricity generation. In the supplemental appendix (subsection E.1), I confirm that these data matches well the annual reporting from the US Energy Information Administration (EIA).

4.2.1 Total Electricity Generation

We start by inspecting how electricity consumption changes following oil price shocks. Figure 10 presents the impulse response functions for total electricity generation and for the ratio of elec-

8. Vietnam is excluded from both the generation and capacity samples, as Ember itself flags its reporting as intermittent and unreliable.

tricity generation to industrial production. The latter is weighted by the previous year’s real GDP to ensure comparability across countries — since industrial production is a country-specific index. Total generation contracts. The drop peaks at -0.57 percentage points. The electricity-to-industrial production ratio confirms that electricity efficiency — the inverse of our ratio, output per TWh — does not increase. In particular, it does not change significantly throughout. As a result, we find no evidence that electricity steps in for other sources of secondary energy.

4.2.2 Clean energy sources

Figure 11 plots the impulse response functions for clean electricity generation. Total renewable energy generated decreases roughly a year after the shock by 1.63 percentage points and stabilizes at 0.51 percentage points at a 2-year horizon.

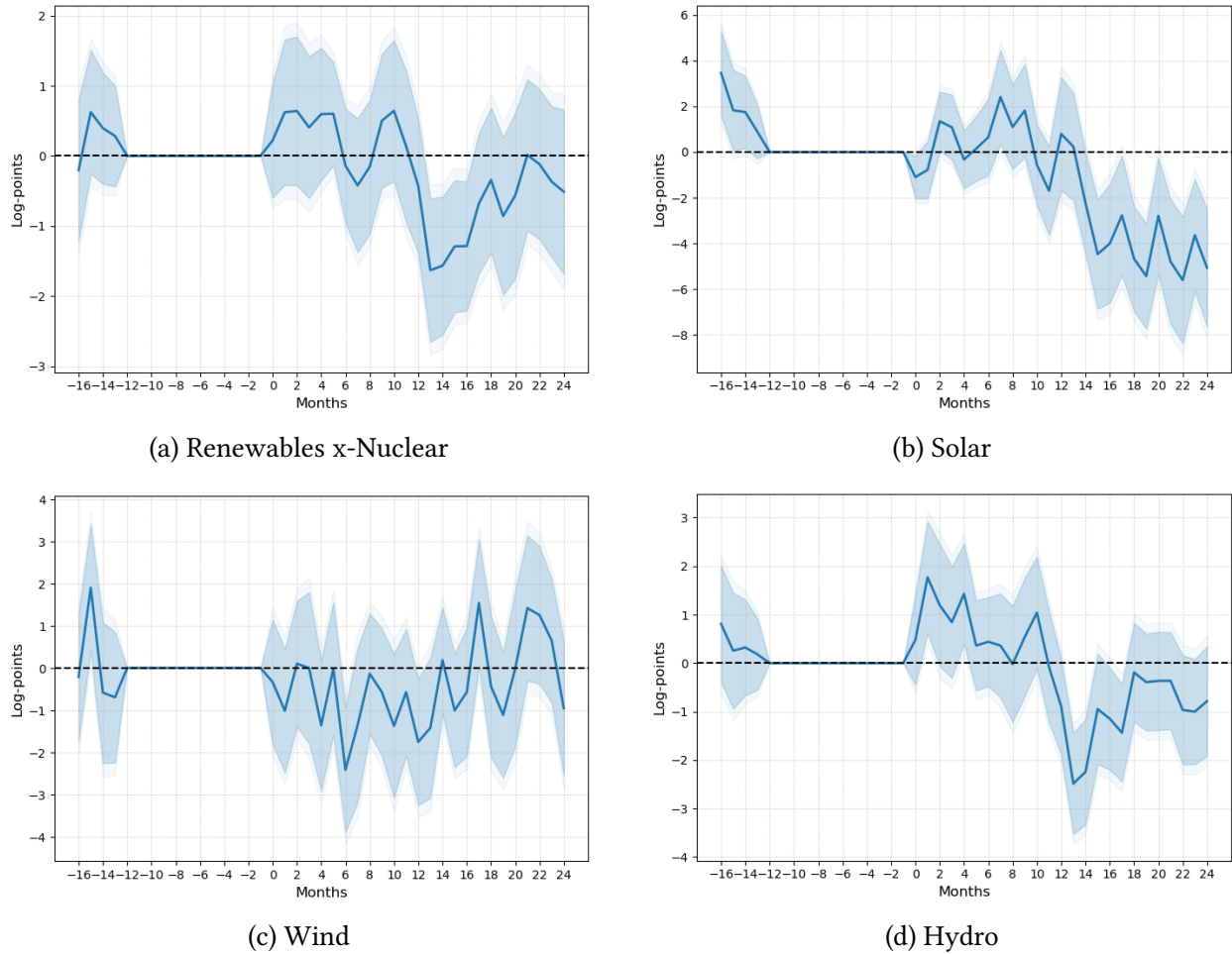
This result appears driven almost entirely by the decline in solar generation which collapses after 1 year of steady production, standing 5.06 percentage points below its trend after 2 years. We show in the supplemental appendix subsection F.2 that this decline surpasses the total electricity fall, significantly as we move towards the full 2-year horizon. In the supplemental appendix (subsection F.3), we observe very similar dynamics for solar nameplate capacity, with the difference that the decline relative to trend occurs much faster — potentially due to the much shorter and narrower panel.

In contrast, wind drops slightly in the first 6 months but the variation in the point estimates points to noise as opposed to a proper steady decline. Hydro in turn actually increases in the first 6 months, peaking at 1.77 percentage points but this increase is short-lived, and it actually also declines 1 year after the increase in oil prices, by 0.90 percentage points, before rebounding.

4.2.3 The Drivers of Solar’s Decline

The results for solar take-up reflect three compounding mechanisms. First, solar power is intermittent — it generates only when irradiance is sufficient — and therefore requires fast-reacting dispatchable backup capacity, primarily natural gas peaker plants, to balance the grid during periods of low generation (Verdolini, Vona, and Popp 2018; Stöckl and Zerrahn 2023). When oil shocks simultaneously disrupt natural gas supply and raise gas prices, the effective cost of deploying complementary solar capacity rises, dampening new investment. Second, the inflationary nature of the shock triggers an endogenous tightening of monetary policy (Bernanke, Gertler, and Watson 1997), causing credit conditions to tighten — an environment that does not support capital-intensive activities such as solar installation (Bistline, Mehrotra, and Wolfram 2023; Aghion et al. 2024; Ameli et al. 2021; Chen and Lin 2024). Third, and separately, macroeconomic and geopolitical uncertainty rise sharply following the shock. Unlike the credit channel,

Figure 11: Response of Renewable Electricity Generation by Source to an Oil Supply News Shock (TWh)



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly electricity generation (TWh) from the respective source. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective generation series (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

elevated uncertainty is a distinct deterrent to long-lived, high fixed-cost investments, by raising the option value of waiting (Bloom 2009).

Aggregate local projections confirm the latter two channels. Figure 12 shows that the 10-year nominal interest rate, obtained from FRED, rises sharply on impact, reaching 0.14 percentage points after 2 months. The term spread initially rises as long-term rates react before short-term rates catch up with monetary policy tightening, after which the spread narrows. Macroeconomic uncertainty (Jurado, Ludvigson, and Ng 2015) spikes on impact and remains elevated, and geopolitical risk (Caldara and Iacoviello 2022) rises persistently — all consistent with the suppression of renewable capacity additions documented in panel b) of figure 11.

5 Conclusion

This paper documents the global environmental consequences of increases in oil prices. The aggregate picture is one of net improvement: a 10% increase in oil prices reduces atmospheric CO_2e concentration persistently. This decline represents only 3.24% of average yearly addition of CO_2e over our sample.

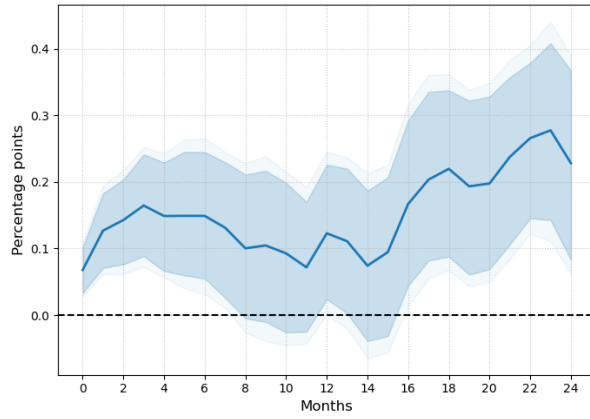
Notwithstanding, the underlying dynamics of individual greenhouse gases reveal important countervailing forces. At the fossil fuel level, the response is heterogeneous — with retail fuels falling faster and more persistently than commercial hydrocarbons such as jet kerosene or bunker fuel — but the decline is general. Natural gas consumption falls, reflecting supply disruption rather than demand substitution. On the contrary, coal exhibits a bifurcated response: steam and coking coal production decline alongside economic activity, while lignite — the cheapest and most CO_2 -intensive grade — rises. Bioenergy generation trends upwards as well. A consequence of this is that N_2O 's concentration in the atmosphere, a potent and long-lived greenhouse gas, rises persistently.

The clean-energy transition does not speed up as a result of oil price increases. ICE registrations fall persistently and remain depressed for most of the projection horizon. Similarly, non-ICE vehicle demand declines but rebounds faster. This is a temporary trend and does not represent an acceleration in alternative vehicle take-up, with the percentage of non-ICE vehicle registrations statistically indistinguishable from its trend.

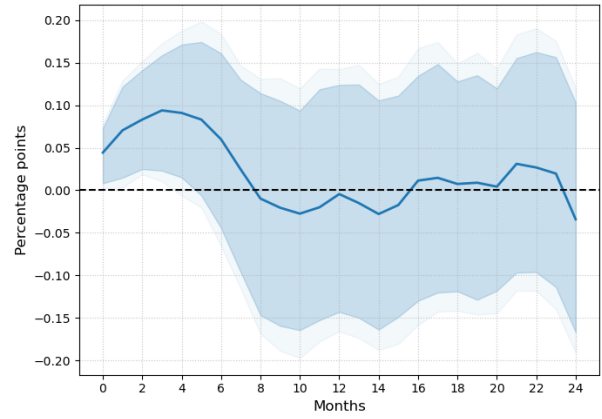
In the power sector, the story is similar. Total electricity generation falls by roughly the same amount as economic activity. Solar generation and installed capacity contract, driven by the combination of monetary-policy tightening that raises the cost of capital, rising geopolitical uncertainty, and grid-level complementarities between intermittent solar and dispatchable natural gas — whose supply is itself disrupted by the shock.

As a result, the emissions efficiency improvement observable in the very short term — emissions per unit of industrial production decline in the second quarter following the shock — is

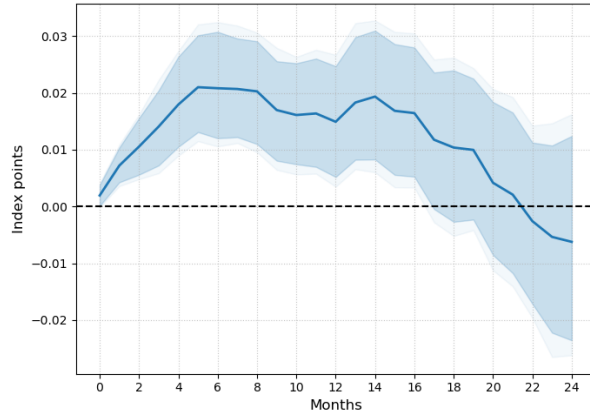
Figure 12: Financial Barriers to Clean Energy Adoption Following an Oil Supply News Shock



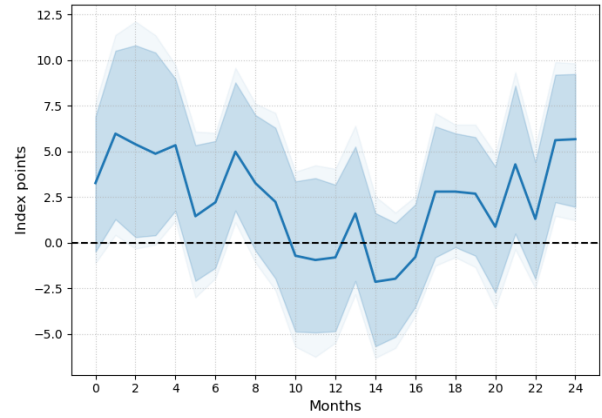
(a) 10-Year Nominal Interest Rate



(b) 10-Year – 1-Year Nominal Term Spread



(c) Macroeconomic Uncertainty (Jurado–Ludvigson–Ng)



(d) Geopolitical Risk

Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable in each panel is the $h + 1$ long-difference in the respective series. Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

accordingly a short-lived intensive-margin response to higher energy costs, not a structural decoupling of output from emissions. It resembles the cyclical covariation between activity and carbon documented by Heutel (2012) and Doda (2014) more closely than the permanent technological decoupling achievable through directed energy innovation (Hassler, Krusell, and Olovsson 2021; Känzig and Williamson 2024).

These findings carry implications for how oil market disruptions should be interpreted through a climate lens. An oil price shock delivers a partial and distorted carbon price signal: it compresses activity in oil-intensive transport but simultaneously undermines capital-intensive renewable investment and stimulates low-grade combustion alternatives at the dispatchable margin.

Focusing on the green transition in the long-run, future work should study how uncertainty in oil and gas production and the subsequent volatility in prices promotes the transition to green energy. This mechanism is commonly alluded to in discussions surrounding oil disruptions, but is not observed in my short-run analysis, in part because other sources of uncertainty and financial constraints overwhelmingly affect the installation and development of clean energy alternatives.

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